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TaIncBC: Topic-aware In-context Prompt with Bias Calibration for Event Causality Identification

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Abstract

Event Causality Identification (ECI) is to determine whether there exists a causal relation between two event mentions. Recently, the prompt learning paradigm has been applied in the ECI task to leverage a large-scale pre-trained language model through carefully designed prompts, yet their contexts do not describe causal relations. In this paper, we propose to include some topic-aware demonstration samples with their ground-truth labels into the query prompt to form an in-context input prompt, so as to facilitate PLM's understanding and comprehending of universal causal relations. Although including demo samples into prompts may enrich in-context causality information, care must be taken for potential bias that could also be introduced into the event prediction model by such demo samples. To address this challenge, we further design a nearest neighbour-based bias calibration to help alleviating demonstration deviations for our topic-aware in-context prompt learning. Our model, called TaIncBC, is experimented on the widely used EventStoryLine and Causal-TimeBank corpus and results validate our design objective in terms of significantly performance improvements over the state-of-the-art algorithms.

Keywords: Event causality identification, In-context Learning, Topic-aware retrieval, Bias calibration, Large-scale Language model

1 Introduction

Event Causality Identification (ECI) aims at detecting whether there exists a causal relation between two event mentions in a document, which is of great importance for many Natural Language Processing (NLP) applications, such as question answer (Bondarenko et al, 2022; Sui et al, 2022), information extraction (Xiang and Wang, 2019), event-centric knowledge graph construction (Rospocher et al, 2016; Guan et al, 2023), and etc. It also has many applications in real world scenarios, such as event prediction (Preethi et al, 2015), strategy optimization (Balgi et al, 2022), and etc. Fig. 1 illustrates an example of causality annotations between event mentions in an accident topic document from the Event StoryLine Corpus (ESC). Fourteen event pairs are annotated with causal relations, of which five are intra-sentence causalities and nine are cross-sentence causalities.



Fig. 1: Illustration of event causality annotation for an accident topic document in the ESC corpus. Event mentions are annotated as one or more words in a raw sentence, and causal relation annotations can exist in intra-sentence or cross-sentence event mentions.

Some graph neural networks have been designed to learn the representation of each event from various semantic connections in a dcoument for relation classification, such as the Rich Graph Convolutional Network (RichGCN) (Phu and Nguyen, 2021), Event Relational Graph Transformer (ERGO) (Chen et al, 2022), Graph-based Event Structure Induction model (Fan et al, 2022). However, they have ignored to utilize some common sense knowledge to augment causality identification. Some researches obtain causal knowledge from relational knowledge bases, like ConceptNet (Speer et al, 2017), FrameNet (Baker et al, 1998), VerbNet (Schuler, 2006), for augmenting causality identification (Liu et al, 2020; Cao et al, 2021; Zuo et al, 2021a, 2020). A triplet in such relational knowledge bases, which contains two concept nodes with a semantic relation between them, can be used to boost the events' representation learning.

The recent prompt learning paradigm (Liu et al, 2021) designs appropriate natural language prompts to reformulate downstream task, so as to utilizing the pre-trained language model (PLM) to predict the desired answer word. Shen et al (2022) have successfully applied a kind of derivative prompt joint learning in the ECI task by designing two relevant prompt tasks to enhance the capability of a PLM for causality prediction. Although their complementary prompts are derived from the ECI task,

they have different optimization objectives with the main ECI task, as their contexts do not describe causal relations but predicting answer words for some other relevant tasks. The in-context learning paradigm (Dong et al, 2023) has just been proposed to include some demonstration samples with their ground-truth labels into the query prompt to form an in-context input prompt, which helps the query prompt to learn some patterns hidden in demonstrations when making its prediction.

We argue that including some topic-aware retrieved demo samples can facilitates PLM's understanding and comprehending of universal causal relations for in-context based event causality identification. On the one hand, the non-causal relation demo samples with the same topic as that of the query instance can train the PLM to be capable of distinguishing whether there exists a causal relation between similar topic events; On the other hand, the causal relation demo samples with different topic as that of the query instance can enhance the generalization ability of PLM to recognize event causalities in different topic documents. This is also in accordance with the philosophy of the contrastive learning to some extent, that is, differentiating the negative (non-causal) samples and intergrating the positive (causal) samples (Jaiswal et al, 2021).

However, the demonstrations could also introduce some potential bias into the PLM (Li et al, 2023a), which may cause unexpected performance degradation of the topic-aware in-context learning model. Considering that similar in-context prompt inputs may have approximative prediction bias, the annotated labels of most similar in-context prompts in the training dataset can be adapted to help alleviating demonstration deviations. This paper also explores how to reduce the deviations introduced from retrieved demonstration samples by nearest neighbour-based bias calibration.

In this paper, we develop a topic-aware in-context learning framework with bias calibration, named TaIncBC, to identify event causalities. Specifically, we first include some topic-aware demonstration samples into a simple cloze style prompt to predict causality virtual answer word. To enhance model prediction via demonstrations, we retrieve negative samples from those annotated gold documents with the same topic as that of the query instance; While the positive samples are retrieved from gold documents with other topics. Besides, we design a nearest neighbours-based bias calibration method by adapting the annotated labels of most similar in-context prompt predictions in the training dataset to help alleviating demonstration deviations. Experiments results have shown that our proposed method significantly outperforms the state-of-the-art algorithms on the ESC and CTB corpus [1].

2 Related Work

2.1 Event Causality Identification

The ECI task is usually approached as a graph-based node classification or edge prediction problem (Phu and Nguyen, 2021; Cao et al, 2021; Fan et al, 2022; Chen et al, 2022). For example, Phu and Nguyen (2021) construct a graph by modelling diverse connections in between words of the document, like position connection, syntactic dependency and etc., and use a graph convolutional network to learn the representation

[1] Source codes will be released after the anonymous review.

of event mention nodes for causality identification. Fan et al (2022) build an event co-reference and co-occurrence graph to identify event causal relation with a graph convolutional network.

Some studies try to discover the causal patterns from external structural knowledge bases to implement a kind of knowledgeable event causality inference (Zuo et al, 2020, 2021b). For example, Zuo et al (2020, 2021b) directly exploit them for data expansion and augmentation in model training. Liu et al (2020) propose to mine a kind of event-agnostic and context-specific patterns from the ConceptNet to enhance the ability of their model for previously unseen cases. Cao et al (2021) encode some graph-structured knowledge from the ConceptNet, including descriptive graph knowledge and relational path knowledge to augment event causality reasoning.

A few studies have applied the prompt learning via designing appropriate prompt templates (Liu et al, 2020; Shen et al, 2022). For example, Liu et al (2020) use an event mention mask generalization mechanism to encode some event causality pattern. Shen et al (2022) use a masked language model as main prompt to predict the causality between event pair. They further design two derivative prompt tasks to leverage potential causal knowledge in PLM for explicit causality identification based on the causal cue word detection.

2.2 In-context Learning and Bias Calibration

After the emergence of large-scale language models, the prompt learning has become a new paradigm for many NLP tasks (Xiang et al, 2022; Seoh et al, 2021; Wang et al, 2021). The in-context learning further improves it by including few in-context demo samples into prompts to enrich context information for analogy learning (Rubin et al, 2022; Xie et al, 2022; Hu et al, 2022; Wu et al, 2023). For example, Hu et al (2022) reformulate dialogue state tacking task into a text-to-SQL problem and propose to retrieve annotated dialogues as demo samples when applying in-context learning to the task. Zhang et al (2022) formulate the demo selection task of in-context learning as a sequential decision problem and propose a reinforcement learning algorithm for identifying generalizable policies to select demo samples. Wu et al (2023) propose a self-adaptive mechanism to select and permutation demo samples.

A few recent studies have noticed the model bias problem that could be introduced by prompt templates and/or demo samples in the in-context learning (Si et al, 2023; Zhao et al, 2021; Han et al, 2022; Li et al, 2023b). For example, Zhao et al (2021) estimate the model bias by using a content-free test input to ask prediction, and fit the calibration parameters to the actual prediction. Han et al (2022) propose to learn a better decision boundary for a language model based classification by estimating the clusters of all categories. Li et al (2023b) propose to alleviate in-context deviations by transforming the PLM output encodings.

To the best of our knowledge, the in-context learning has not yet been tried for the ECI task. The proposed TaIncBC is such a trial which retrieves topic-aware demo samples and designs nearest neighbour-based bias calibration for the ECI task.

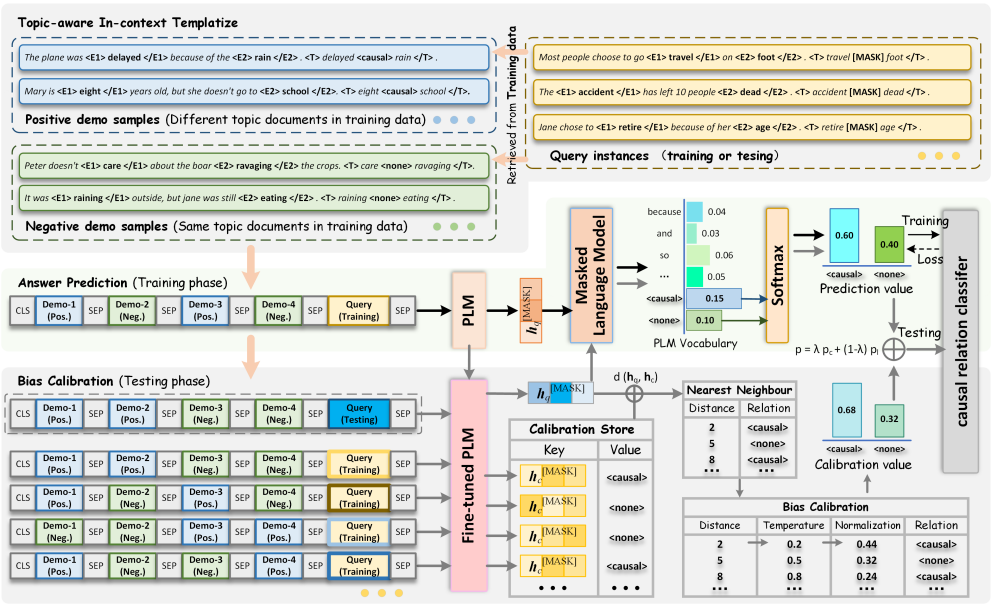


Fig. 2: Illustration of our TaIncBC framework.

3 Method

As illustrated in Fig. 2 our TaIncBC has three main processes, including the in-context prompt templatize, answer prediction and bias calibration.

3.1 Demonstration Retrieval

The in-context input prompt contains a query instance for answer prediction and a few demonstration samples for contextual analogy. The query instance q is an event pair, i.e., $q = (Evt_i, Evt_j)$, with the causal relation between the two events to be identified. For q, we first retrieve some demonstration sample $d_k = (Evt_m, Evt_n, y)$ from the training dataset with its annotated label $y \in \{causal, none\}$.

A negative demo sample d_k^- is defined as the event pair with a *none* label, which is retrieved from the annotated gold documents with the same topic as that of the query instance. A positive demo sample d_k^+ is defined as the event pair with a *causal* label, which is retrieved from gold documents with topics different from the query instance. By using both positive and negative demonstrations, we argue that the TaIncBC model can be trained to be capable of distinguishing non-causal relation from the same topic and identifying causal relation from different topics. As we have no prior about which kind is more beneficial, we retrieve the same number of positive and negative demo samples, yet the demo number will be examined via experiments.

3.2 Prompt Templatize

We reformulate the query instance q and its retrieved demo sample d_k into the prompt template $T(q)$ and $T(d_k)$, respectively, as illustrated in the top of Fig. 2. We use a simple template for the query instance:

$$T(q) = Evt_i \texttt{ [MASK] } Evt_j$$

The Evt_i and Evt_j are the two event mentions of the query instance, and the PLM-specific token `[MASK]` is inserted between them for relation prediction. For demo samples, we use the same template of the query instance, but replacing the PLM-specific token `[MASK]` by a virtual word `<causal>` for each positive sample d_k^+ or `<none>` for each negative sample d_k^-:

$$T(d_k) = \begin{cases} Evt_m \texttt{ <causal> } Evt_n, & \text{if } d_k \text{ is } d_k^+, \\ Evt_m \texttt{ <none> } Evt_n, & \text{if } d_k \text{ is } d_k^-. \end{cases}$$

As an event mention is often annotated by a few words in a sentence, we concatenate the raw sentences of event mentions $T(s)$ with the prompt template of a query instance $T(q)$ and those templates of demo samples $T(d_k)$, so as to including their full contextual semantics. Furthermore, we design a few special tokens, including `<E1>`, `</E1>`, `<E2>`, `</E2>`, `<T>` and `</T>`, to indicate the beginning and ending of event mentions and prompt template. Finally, we concatenate the query prompt template and those prompt templates of its retrieved demonstrations as the input to a PLM. While the PLM-specific tokens `[CLS]` and `[SEP]` are used to indicate the beginning and ending of an input, and some `[SEP]` tokens are used as separators between the query and those demonstrations.

3.3 Answer Prediction

After the PLM encoding, we obtain a hidden state $\mathbf{h}_q$ for the input `[MASK]` token for causality prediction, where $\mathbf{h}_q \in \mathbb{R}^s$ and s is the dimension of the hidden state. The Masked Language Model (MLM) classifier of the PLM uses the hidden state $\mathbf{h}_q$ to estimate the probability of each word in its vocabulary V for the `[MASK]` token as follows:

$$P(\texttt{[MASK]} = v \in V \mid T(s)). \tag{1}$$

We add two virtual answer words into a PLM's vocabulary to represent whether a causal relation exists, viz. the `<causal>` and `<none>`. Then a softmax layer is applied on the prediction scores of the two virtual answer words to normalize them into probabilities:

$$P_l(v_i \in V_a \mid T(s)) = \frac{e^{p_{v_i}}}{\sum_{j=1}^{n} e^{p_{v_j}}}, \tag{2}$$

where $V_a = \{\texttt{<causal>}, \texttt{<none>}\}$.

3.4 Training strategy

In the training phase, we tune the PLM parameters as well as the MLM classifier parameters based on the in-context prompt template and newly added virtual answer words. We adopt the cross entropy loss as the cost function:

$$J(\theta) = -\frac{1}{T}\sum_{t=1}^{T}\mathbf{y}^{(t)}\log(\hat{\mathbf{y}}^{(t)}) + \lambda\|\theta\|^2, \tag{3}$$

where $\mathbf{y}^{(t)}$ and $\hat{\mathbf{y}}^{(t)}$ are the gold label and predicted label for the t-th query instance, respectively. λ and θ are the regularization hyper-parameters. We use the AdamW optimizer Loshchilov and Hutter (2019) with $L2$ regularization for model training.

3.5 Bias Calibration

To alleviate potential bias introduced by demo samples, we design a nearest neighbour-based bias calibration for the in-context prompt prediction, as illustrated in the bottom of Fig. 2.

After model training, we again input the in-context prompt templates for each instance in the training dataset $\mathcal{D}_{train}$ into the fine-tuned PLM to construct a calibration store $\mathcal{S}$, which consists of some key-value pairs of the obtained `[MASK]` hidden state $\mathbf{h}_c \in \mathbb{R}^s$ and annotated relation label y_c:

$$\mathcal{S}(c) = \{(\mathbf{h}_c, y_c), \forall x_c \in \mathcal{D}_{train}\}. \tag{4}$$

In the testing phase, we first retrieve some nearest neighbours in the calibration store for the test instance based on the Euclidean distance $\mathsf{d}(\mathbf{h}_q, \mathbf{h}_c)$ between its encoded `[MASK]` hidden state $\mathbf{h}_q$ and that of calibration samples $\mathbf{h}_c$. These nearest neighbours form a set $\mathcal{N}_t \subseteq \mathcal{S}$. We next apply a temperature T (greater than one) on the distance d for each $\mathbf{h}_c \in \mathcal{N}_t$ to flatten its distribution. The nearest neighbours with their annotated labels are used to compute the calibration score distribution, as follows:

$$P_c(y|\ \mathbf{h}_q) = \sum_{(\mathbf{h}_c, y_c)\in\mathcal{N}_t} \mathbb{1}_{y_c=v_a} \cdot \text{softmax}(\frac{-\mathsf{d}(\mathbf{h}_q, \mathbf{h}_c)}{\mathsf{T}}),$$

where softmax$(\cdot)$ is for normalization.

3.6 Causality Identification

The final event causality prediction of our TaIncBC is computed by both the in-context prompt prediction distribution P_l and the calibration score distribution P_c, as follows:

$$p(y|\ x) = (1-\lambda)\cdot p_l + \lambda \cdot p_c, \tag{5}$$

where $\lambda \in [0, 1]$ is a weight coefficient to balance the importance of in-context prompt prediction and bias calibration. When $p(causal)$>$p(none)$, we accept that there exists a causal relation; Otherwise, there exists non-causal relation.

4 Experiment Settings

In this section, we present our experiment settings, including the dataset, PLMs, competitors, and parameter settings.

4.1 Datasets

Our experiments are conducted on two widely used datasets for the ECI task, that is, the EventStoryLine 0.9 Corpus (ESC) Caselli and Vossen (2017) and the Causal-TimeBank Corpus (CTB) Mirza and Tonelli (2014).

EventStoryLine contains 22 topics and 258 documents from various news websites. There are in total 5,334 event mentions in the ECS dataset. A total number of 5,625 event pairs are annotated with causal relations, among which 1,770 causal relations are from intra-sentence event pairs and 3,855 causal relations are from cross-sentence event pairs. Following the standard data splitting Gao et al (2019), we use the last two topics as the development set, and conduct 5-fold cross-validation on the remaining 20 topics. The average results of precision, recall, and F1 score are adopted as performance metrics.

Causal-TimeBank contains 184 documents from English news articles and 7,608 annotated event pairs. A total number of 318 event pairs are annotated with causal relations, among which 300 causal relations are from intra-sentence event pairs and only 18 causal relations are from cross-sentence event pairs. Following the standard data splitting Liu et al (2020), we employ a 10-fold cross-validation and the average results of precision, recall, and F1 score are adopted as performance metrics. Following Phu and Nguyen (2021), we only conduct intra-sentence event causality identification experiments on CTB, as the number of cross-sentence event causal pairs is quite small.

4.2 Competitors

We compare our TaIncBC with the following competitors:

ILP Gao et al (2019) uses integer linear programming to identify causal relations based on document-level causal structure constraints.

KnowDis Zuo et al (2020) uses external lexicon knowledge to distantly augment causality data.

KnowMMR Liu et al (2020) uses external knowledge to mine some event causality patterns.

CauSeRL Zuo et al (2021a) adopts a contrastive strategy to transfer learned external causal statements.

LearnDA Zuo et al (2021b) leverages knowledge bases to interactively generate training data.

LSIN Cao et al (2021) uses graph induction to learn external structural and relational knowledge.

RichGCN Phu and Nguyen (2021) uses a graph convolutional network to learn document context-augmented representation for event pairs.

DPJL Shen et al (2022) leverages two derivative prompt tasks to identify causality.

GESI Fan et al (2022) designs a graph convolutional network on an event co-reference graph.

ERGO Chen et al (2022) builds an relational graph using event pairs as nodes to capture causation transitivity via a transformer-like network.

SemSIn Hu et al (2023) uses a GNN to learn from event-centric structures for encoding events.

CHEER Chen et al (2023) builds a heterogeneous event interaction graph to integrate event centrality information for causality reasoning.

SENDIR Yuan et al (2023) constitutes a reasoning chain based on intra-sentential event pairs for inter-sentential event causality reasoning.

4.3 Parameter Setting

We use the pre-trained RoBERTa Liu et al (2019) with its 768-dimension base version provided by the HuggingFace transformers[2] Wolf et al (2020). We run PyTorch [3] framework with CUDA on NVIDIA GTX 3090 GPUs. The learning rate is set to 1e-5, the batch size to 16. All trainable parameters are randomly initialized from normal distributions. The number of demo samples is set to 4 in random order, of which 2 are positive and 2 negative. As the positive and negative demo samples are unbalanced in the training dataset, we randomly sample 10^3 positives training samples and 10^3 negatives training samples for the calibration store construction. We retrieve the top-10 nearest neighbours for bias calibration, and the Temperature T is set to 3, and $\lambda = 0.5$.

Table 1: Comparison of intra-sentence prediction results on the ESC and CTB corpus.

Model	Causal-TimeBank			EventStoryLine		
	P	R	F1	P	R	F1
KnowDis (COLING, 2020)	42.3	60.5	49.8	39.7	66.5	49.7
KnowMMR (IJCAI, 2020)	36.6	55.6	44.1	41.9	62.5	50.1
CauSeRL (ACL, 2021)	43.6	68.1	53.2	41.9	69.0	52.1
LSIN (ACL, 2021)	51.5	56.2	53.7	47.9	58.1	52.5
LearnDA (ACL, 2021)	41.9	68.0	51.9	42.2	69.8	52.6
RichGCN (NAACL, 2021)	39.7	56.5	46.7	49.2	63.0	55.2
ERGO (COLING, 2022)	62.1	61.3	61.7	57.5	72.0	63.9
DPJL (COLING, 2022)	63.6	66.7	64.6	65.3	70.8	67.9
SemSIn (ACL, 2023)	52.3	65.8	58.3	64.2	65.7	64.9
Our TaIncBC	63.5	67.8	**65.2**	66.1	73.5	**69.6**

[2]https://github.com/huggingface/transformers
[3]pytorch.org

Table 2: Comparison of overall prediction results on the ESC corpus.

Model	EventStoryLine								
	Intra-Sentence			Cross-Sentence			Overall		
	P	R	F1	P	R	F1	P	R	F1
ILP (NAACL,2019)	38.8	52.4	44.6	35.1	48.2	40.6	36.2	49.5	41.9
RichGCN (NAACL,2021)	49.2	63.0	55.2	39.2	45.7	42.2	42.6	51.3	46.6
GESI (SIGIR,2022)	-	-	50.3	-	-	49.3	-	-	49.4
ERGO (COLING,2022)	57.5	72.0	63.9	51.6	43.3	47.1	48.6	53.4	50.9
CHEER (ACL, 2023)	56.9	69.6	62.6	45.2	52.1	48.4	49.7	53.3	51.4
SENDIR (ACL, 2023)	65.8	66.7	66.2	33.0	90.0	48.3	37.8	82.8	51.9
Our TaIncBC	66.1	73.5	**69.6**	57.6	59.0	**58.3**	60.4	63.5	**61.9**

5 Results and Analysis

5.1 Overall Results

Table 2 and Table 1 compare the overall performance between our TaIncBC and the competitors on the ESC corpus and CTB corpus. The competitors in Table 1 have only reported the intra-sentence results on the ESC dataset and CTB dataset; While only the competitors in Table 2 have reported all intra-sentence, cross-sentence, and overall results on the ESC dataset.

From Table 1, we can first observe that the DPJL adopting a kind of derivative prompt learning can significantly outperform the other competitors in intra-sentence causality identification. The outstanding performance can be attributed to its applying the prompt learning paradigm that transforms the ECI task to directly predict a PLM vocabulary word, other than fine-tuning a downstream task-specific neural model upon a PLM. Although the other competitors have used some kind of extra knowledge, such as lexicon knowledge and relational knowledge, from large-scale external knowledge bases, the prompt learning model can better enjoy the encyclopedic linguistic knowledge embedded in a PLM during the model training. However, the DPJL has used a causal cue word for event causality identification, which does not exist in a cross-sentence event-pair. As such it cannot be applied in the cross-sentence case.

From Table 2, we can observe that the RichGCN, GESI, and ERGO can obviously outperform the ILP in terms of higher F1 score with all intra-sentence, cross-sentence, and overall event causality identification on the ESC dataset. This can be attributed to the use of some graph-based neural networks, operating on a document-level graph structure with large-scale trainable parameters to augment event representation learning. Indeed, graph-based neural networks have been proven to be effective for many NLP tasks Scarselli et al (2009); Piao et al (2022). We can also observe that the improvement of intra-sentence causality identification on both the ESC and CTB dataset is more

significant than that of cross-sentence. This might be attributed to the use of pre-trained language model for event node encoding, which can capture the semantic interaction between two events in a sentence.

Finally, our TaIncBC has achieved significant performance improvements over all competitors in terms of much higher F1 score with all intra-sentence, cross-sentence, and overall event causality identification on both the ESC and CTB dataset. We attribute its outstanding performance to including some demo samples into the prompt learning as well as calibrating demonstration bias for our topic-aware in-context learning, by which our TaIncBC can not only better exploit some in-context information but also alleviate their introduced deviations.

5.2 Ablation Study

To examine the effectiveness of our topic-aware in-context learning and nearest neighbour-based bias calibration, we design the following ablation studies. Table 3 compares the their results.

• Prompt is only the prompt learning model, without in-context demonstration and hence without bias calibration.

• RaInc (Rand) includes some randomly selected demo samples into the input prompt, but without bias calibration.

• TaInc (Topic) includes our retrieved topic-aware demo samples into the input prompt, but without bias calibration.

• RaIncBC (Rand) includes some randomly selected demo samples into the input prompt and it is with bias calibration.

• TaIncBC (Topic) is the full TaIncBC model, including our retrieved topic-aware demo samples and with bias calibration.

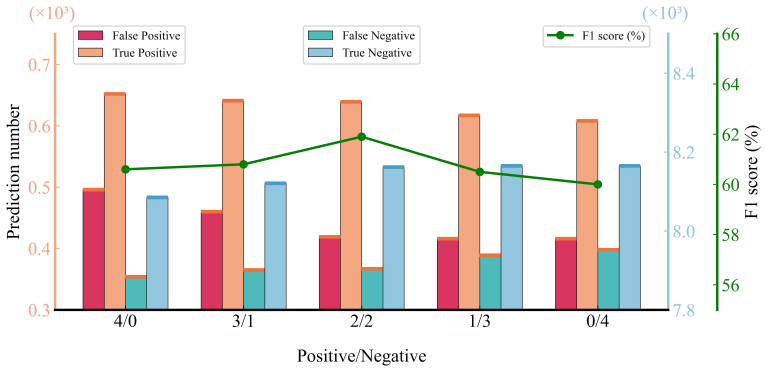
Table 3: Comparison of prediction results of ablation models on the ESC and CTB corpus.

Model	**EventStoryLine**									**Causal-TimeBank**		
	Intra-Sentence			Cross-Sentence			Overall			Intra-Sentence		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Prompt	70.3	62.0	65.2	63.3	47.0	53.0	65.6	51.6	57.0	56.4	54.6	54.8
RaInc (Rand)	65.0	67.9	65.9	58.1	53.4	55.3	60.4	58.0	58.8	58.1	53.9	55.1
TaInc (Topic)	67.0	71.2	69.0	59.2	53.6	56.1	61.9	59.1	60.3	58.7	60.3	58.9
RaIncBC (Rand)	65.4	71.3	68.2	58.0	57.8	57.8	60.4	62.0	61.2	56.6	67.3	61.1
TaIncBC (Topic)	66.1	73.5	**69.6**	57.6	59.0	**58.3**	60.4	63.5	**61.9**	63.5	67.8	**65.2**

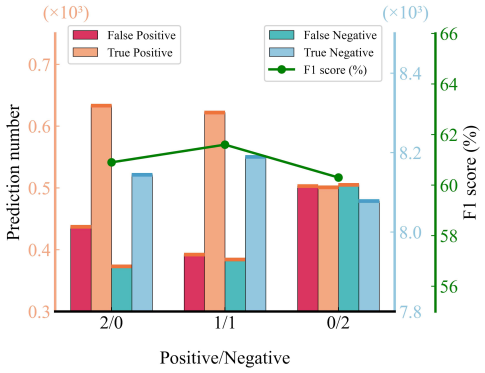
Topic-aware In-context Retrieving: The first observation is that the Prompt cannot outperform the TaInc (Topic) as well as the RaInc (Rand). This suggests the effectiveness of in-context learning for those demo samples enriching contextual analogy, even the demo samples are randomly selected. Besides, we can also observe that the RaInc (Rand) cannot outperform TaInc (Topic) which retrieves topic-aware demo samples other than random selection. This validates one of our design objectives for

enhancing the PLM encoding capability via both positive and negative demonstration. The positive demonstration could help the PLM to understand the causality of other event kinds in different topic documents. The negative demonstration could help the PLM to differentiate similar yet non-causal events in the same topic documents.

Bias calibration: The second observation is that the RaIncBC (Rand) and TaIncBC (Topic) with bias calibration can further improve the performance of RaInc (Rand) and TaInc (Topic), respectively. This suggests that our nearest neighbour-based bias calibration works to alleviate demonstration deviation. Likewise, we can observe that the RaIncBC (Rand) cannot outperform the TaIncBC (Topic), which again validate our design objective of topic-aware demonstration retrieval for augmenting in-context learning.



(a) Demo number 4



(b) Demo number 2

Fig. 3: Comparison of causal predictions when using different numbers of positive and negative demo samples. We use the average predicted numbers and results of 5-fold cross-validation on the ESC corpus for illustration.

Demonstration effectiveness: To further investigate the effect of demonstration for in-context learning, we compare the performance of using different numbers of

retrieved demo samples. Fig. 3 presents the experimentation results, we use the average predicted numbers and results of 5-fold cross-validation on the ESC corpus for illustration. We can observe that the best performance is achieved when using the same number of positive and negative samples. This is not unexpected. The contextual analogy for the TaIncBC model needs to be balanced between the causal and non-causal relation, so as not to favor a particular kind especially when no prior knowledge is available on which kind more beneficial.

Indeed, our analysis is also collaborated by Fig. 3a and Fig. 3b. We can observe that the TaIncBC model with only positive demonstrations (the cases of 4/0 and 2/0 in the figure) have predicted more causal relations than the other cases. This also validates the philosophy of in-context learning, that is, the contextual analogy from demonstrations could guide what a model would be like. In the cases of only positive demonstrations, the trained TaIncBC model could predict more causal relations, however, they are actually false positives. Such increased false positives due to only positive demonstrations obviously are not welcome for the ECI task, as the overall F1 performance also suffers from false positives. Similar results can be observed in the TaIncBC model with only negative demonstrations.

5.3 Visualization

In the testing dataset, we randomly select 10 positive instances each with a `<causal>` relation and 10 negative instances each with a `<none>` causal relation. Fig. 4 visualizes the prediction probabilities (after softmax) of these 20 instances when using different ablation study models. Note that the Topic-aware (reverse) applies a reverse retrieve strategy in our TaIncBC model. That is, it retrieve the positive demo samples from the same topic documents and negative ones from different topic documents.

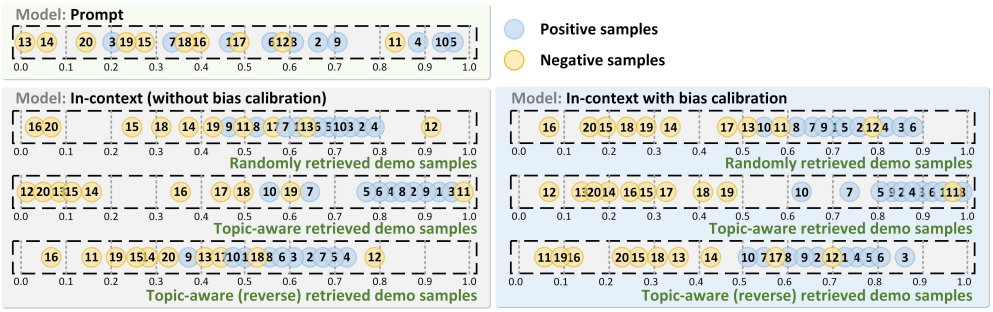


Fig. 4: Visualization of the prediction probabilities of 20 randomly selected test instances for different ablation models and demonstration retrieval strategies.

The first observation is that both the In-context (Topic-aware) and our TaIncBC (Topic-aware) model have corrected many incorrect predictions by the Prompt model, including the positive samples 1, 3, 7 and negative samples 12. This again validates our design objective of applying the in-context learning paradigm with bias calibrations.

The second observation is that different demonstration retrieval strategies have a noticeable impact on in-context learning predictions. For example, the topic-aware based retrieval has corrected the prediction of negative sample 12, but the random and reversed topic-aware missed. Moreover, the random based and reversed topic-aware retrieval have incorrectly changed the prediction of sample 9, 13, 18, etc., but the topic-aware does not. This might be attributed to our designed topic-aware retrieval strategy that enhance the in-context model to identifying causal relation from different topic documents and distinguishing non-causal relation from the same topic.

We can also observe that the in-context with bias calibration can correct most erroneous predictions, like the positive sample 9 in random and negative sample 18 in the reversed topic-aware retrieval. Besides, bias calibration also enhances the probabilities of correctly predicted samples, such as the probability of positive sample 7 is improved from 0.6+ to 0.7+. This indicates the effectiveness of our bias calibration.

6 Conclusion

This is the first paper applying in-context learning paradigm with bias calibration for the ECI task. We have designed a topic-aware demonstration retrieval strategy and reformulate a query instance with its retrieved demo samples into the prompt template for PLM-based causality prediction. The nearest neighbour-based bias calibration method has been developed to help alleviating demonstration deviations for our topic-aware in-context learning. Experiments on the ESC and CTB corpus have validated that our TaIncBC can significantly outperform the state-of-the-art algorithms.

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